

Supplementary Information:

Genome-wide association study in Han Chinese identifies four new susceptibility loci for coronary artery disease

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Supplementary Information

Supplementary Figures

Supplementary Figure 1: Flow chart of two-stage design.

Supplementary Figure 2: Plots of the first two eigenvectors produced by the EIGENSOFT software package in our cohorts and the HapMap samples

Supplementary Figure 3: Q-Q plots of the genome-wide association results

Supplementary Figure 4: The LD structures of the SNPs on 12q24.12-12q24.13 significantly associated with CAD in Chinese Han populations

Supplementary Figure 5: The region plot and LD patterns of the region on 9p21.3 in populations of Chinese and European ancestry

Supplementary Tables

Supplementary Table 1: Characteristics of the overall samples

Supplementary Table 2: Discovery, Replication 1 and Replication 2 results for SNPs selected for replication

Supplementary Table 3: Association results for four previously reported CAD loci in the present GWAS

Supplementary Table 4: Effects of CAD loci on established cardiovascular risk factors

Supplementary Table 5: Derived allele frequency of ethnic-specific association loci on 12q24 in three ethnic groups

Supplementary Table 6: Pairwise linkage disequilibrium coefficients between the four SNPs of 9p21

Supplementary Table 7: Conditional logistic regression analysis for SNPs from 9p21.3

Supplementary Table 8: Comparison of the four new findings between the present GWAS and the CARDIoGRAM datasets

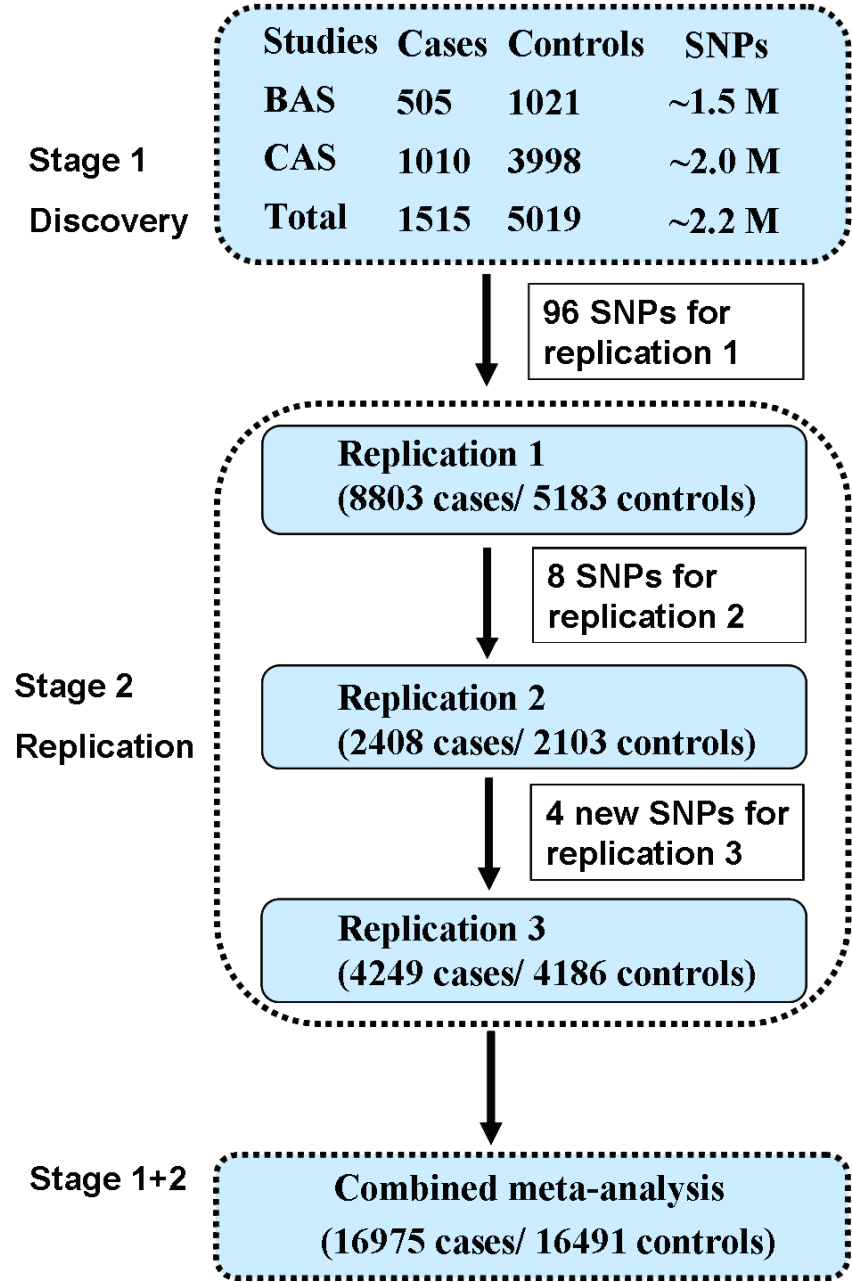
Supplementary Table 9: Comparison of the present GWAS in discovery stage with published GWAS for CAD

Supplementary Note:

The CARDIoGRAM Consortium

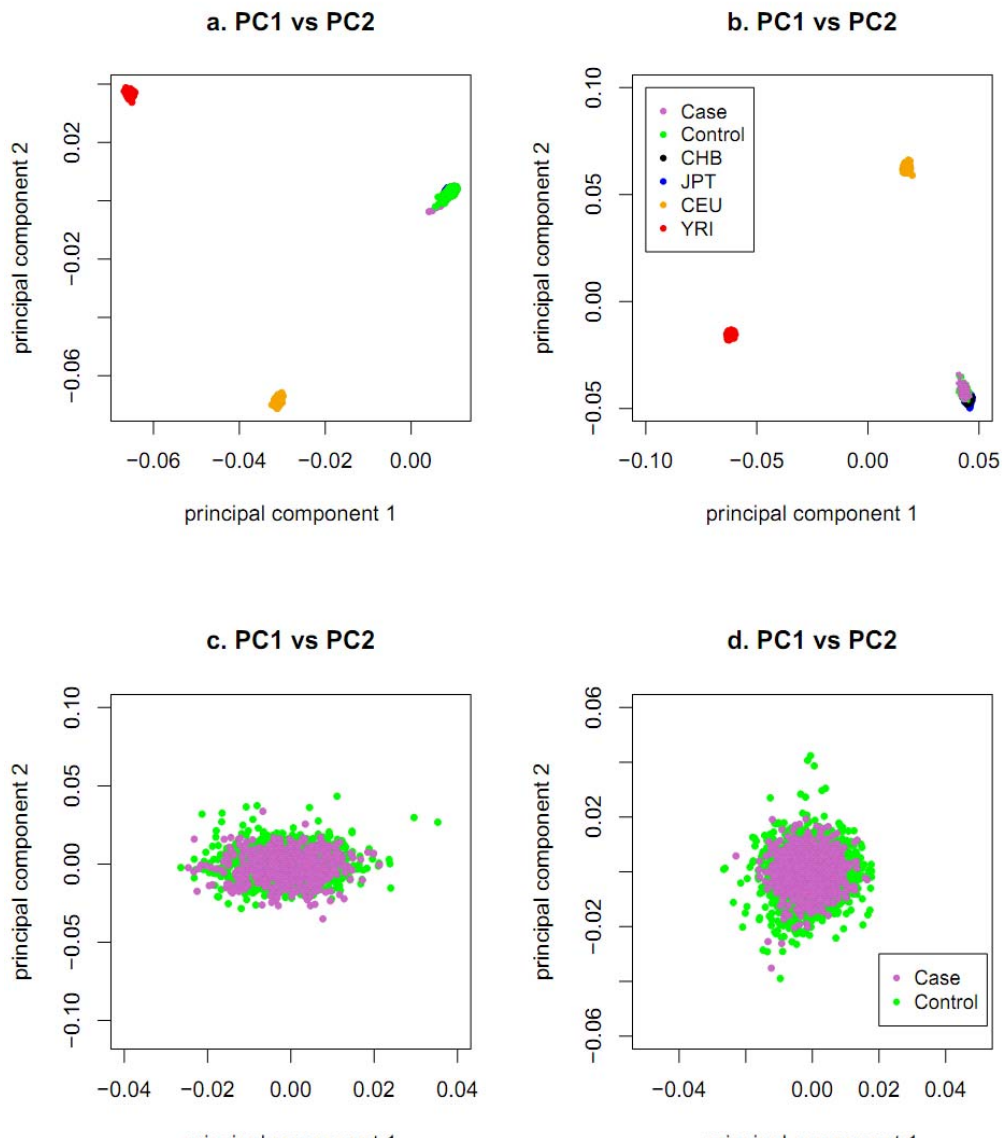
Supplementary References

Supplementary Figure 1: Flow chart of two-stage design



Supplementary Figure 2: Plots of the first two eigenvectors produced by the EIGENSOFT software package in our cohorts and the HapMap samples

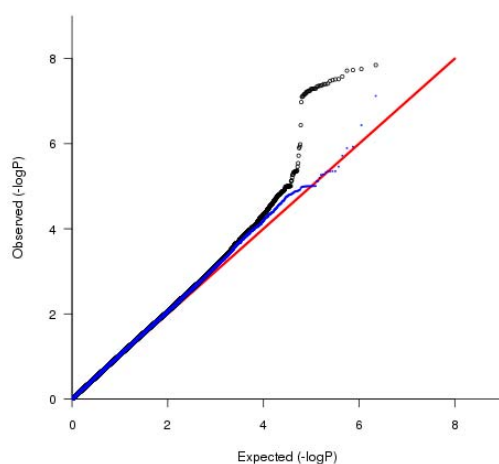
a. BAS individuals (505 MI cases and 1,021 controls) and 618 HapMap 3 Release 3 individuals (including 137 Chinese (CHB), 113 Japanese (JPT), 165 Caucasian (CEU), and 203 Yoruban (YRI)); b. CAS individuals (1010 CAD cases and 3,998 controls) and 618 HapMap samples; c. BAS individuals (505 MI cases and 1,021 controls); d. CAS individuals (1010 CAD cases and 3,998 controls).



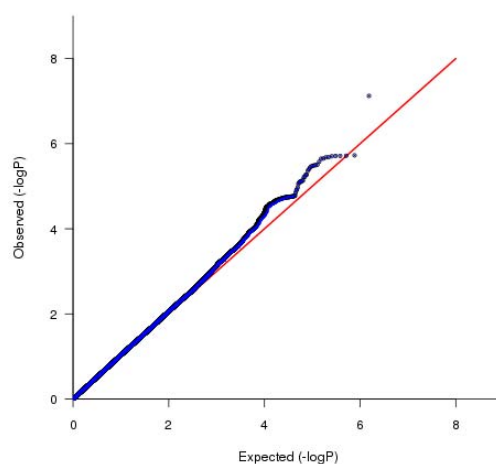
Supplementary Figure 3: Q-Q plots of the genome-wide association results

a. Two studies combined, b. BAS, c. CAS

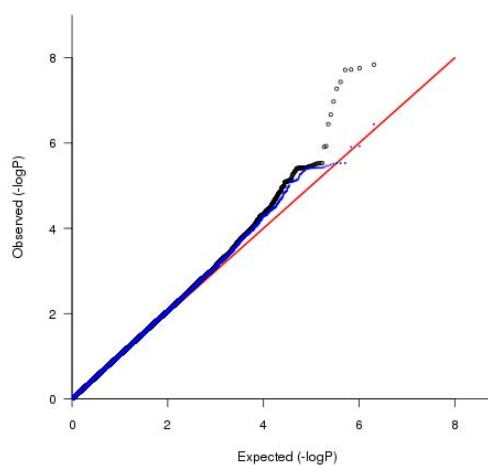
The observed P values (y-axis) were compared with the expected P values under the null distribution (x-axis). The QQ-plot in black includes all SNPs that passed the inclusion criteria, whereas the plot in blue is for the P values after the exclusion of the SNPs with $P < 0.05$ located at 8 loci listed in Table 1 and **Supplementary Table 3**. The equiangular line is included in the plot for reference purpose.



a. Two studies combined $\lambda=1.05$



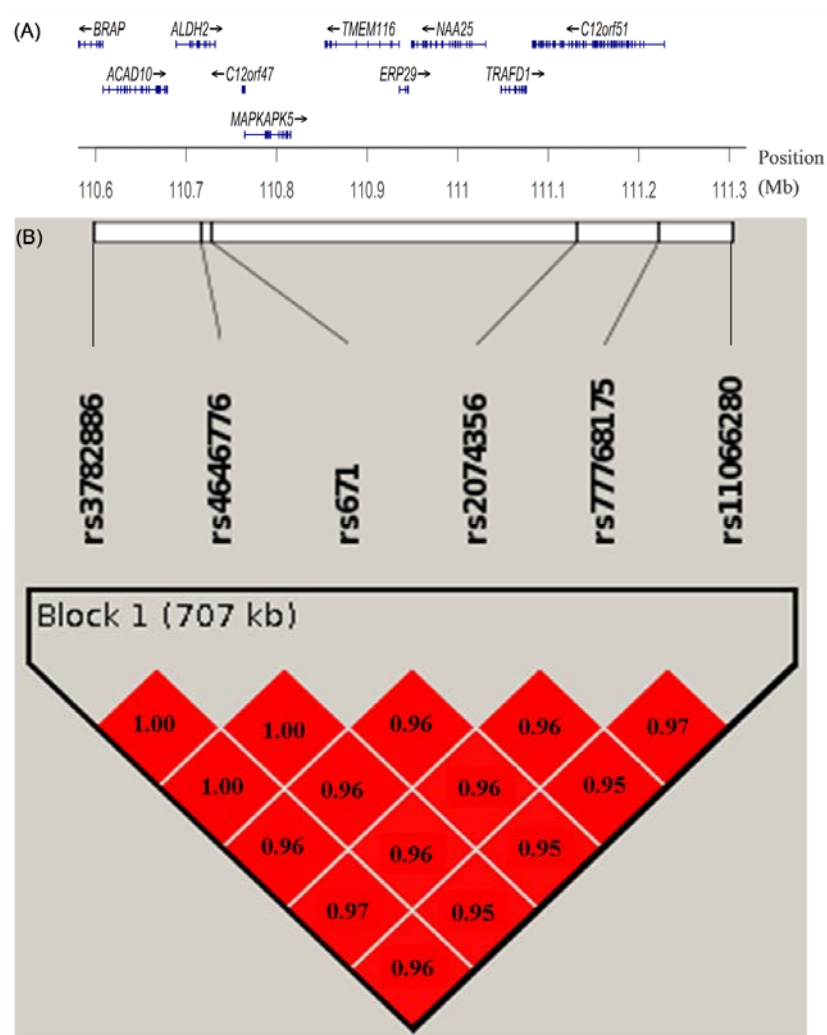
b. BAS $\lambda=1.04$



c. CAS $\lambda=1.04$

Supplementary Figure 4: The LD structures of the SNPs on 12q24.12-12q24.13 significantly associated with CAD in Chinese Han populations

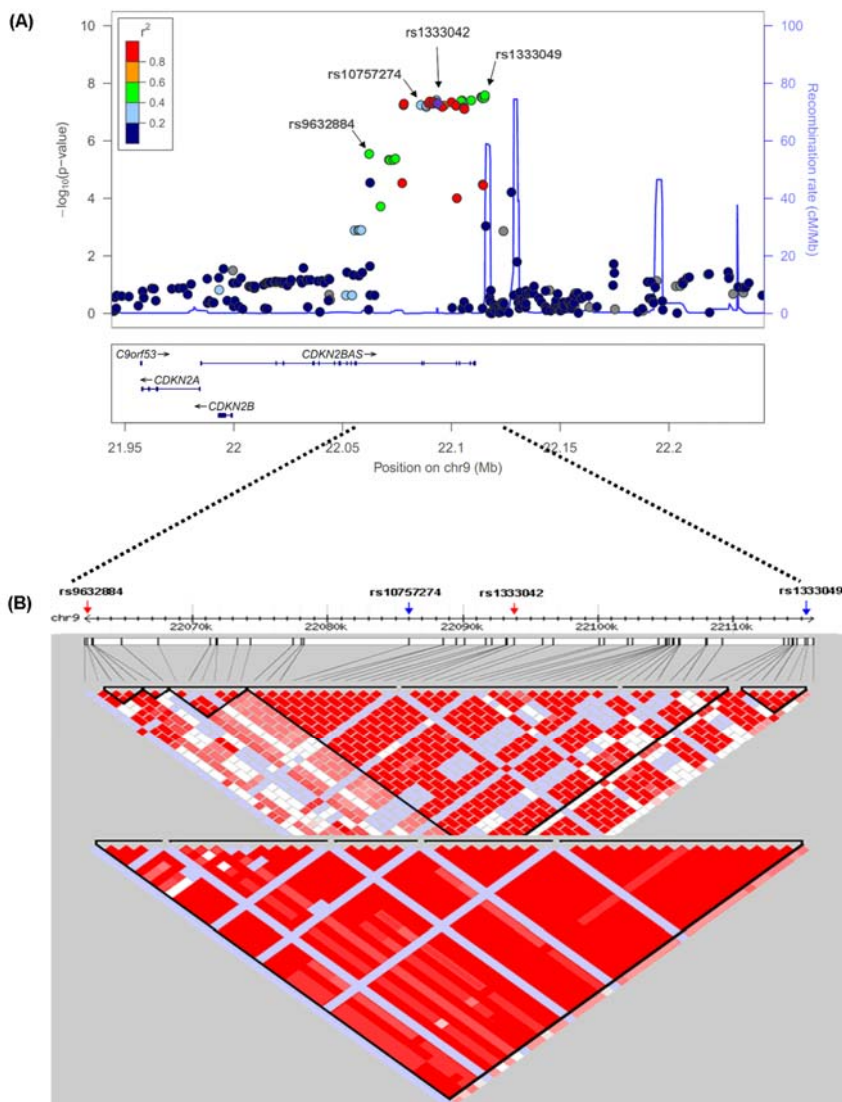
- (A) The top panel shows the known genes within region of interest, approximately 0.7 Mb on 12q24.12-12q24.13. The genes are shown as annotated arrows and the genomic position is based on NCBI build 36.
- (B) The bottom panel indicates that the positions of the six SNPs (rs3782886, rs4646776, rs671, rs2074356, rs77768175, and rs11066280) on 12q24.12-12q24.13 region and the pairwise r^2 values of LD with each other are depicted in the diamonds.



Supplementary Figure 5: The region plot and LD patterns of the region on 9p21.3 in populations of Chinese and European ancestry

Panel (A): Regional association plots for 9p21.3 in our data. SNPs are plotted according to their chromosomal positions (NCBI build36) with their P values (as $-\log_{10} P$ values) of the discovery meta-analysis. The lead SNP (rs1333042) is represented by a purple diamond, and the r^2 values of LD between the lead SNP and the other SNPs are indicated by different colors. The estimated recombination rates from 1000 Genomes June 2010 CHB+JPT samples are plotted in cyan. The genes within region of interest are annotated and are shown as arrows.

Panel (B): The LD patterns (D') were created using the Haploview software and the genotyping data from the HapMap project. The upper panel shows the D' pattern in Chinese population; and the bottom panel shows the D' pattern in the European-descent population. The SNPs (rs10757274 and rs1333049) indicated by blue arrows are in strong LD with each other, are in moderate LD with the two SNPs depicted by red arrows (rs9632884 and rs1333042).



Supplementary Table 1: Characteristics of the overall samples

Stage of study	Discovery-Stage 1				Replication-Stage 2					
	BAS		CAS		Replication 1		Replication 2		Replication 3	
	Cases	Controls	Cases	Controls	Cases	Controls	Cases	Controls	Cases	Controls
sample size	505	1021	1010	3998	8803	5183	2408	2103	4249	4186
Male/Female	430/75	757/264	853/157	1870/2128	7439/1364	4332/851	1905/503	1694/409	3293/956	3072/484
Age at entry into study (yrs)	56.65±8.96	48.19±8.43	50.68±8.16	52.53±7.69	53.65±7.94	57.22±9.09	55.12±7.88	54.76±8.93	57.21±9.76	53.81±8.51
Male	55.78±8.16	47.92±8.41	49.71±7.71	52.67±7.71	52.90±7.90	56.69±8.86	54.22±7.95	54.54±9.01	56.19±9.71	53.42±8.35
Female	61.65±5.60	48.95±8.45	55.92±7.52	52.41±7.66	57.71±6.88	59.94±9.71	58.50±6.59	55.68±8.55	60.75±9.10	56.76±9.12
Age of disease onset (yrs)	52.13±8.62	--	49.18±8.03	--	51.99±7.97	--	54.21±8.04	--	56.16±9.52	--
Male	51.56±8.71	--	48.25±7.77	--	51.18±7.86	--	53.25±8.07	--	55.09±9.41	--
Female	55.43±7.29	--	54.25±7.54	--	56.40±7.11	--	57.84±6.82	--	59.85±8.95	--
PCI(%)	37.00	--	68.91	--	61.29	--	66.49	--	49.47	--
CABG(%)	30.20	--	14.75	--	14.19	--	14.20	--	17.70	--
MI (%)	100.00	0.00	55.54	0.00	74.44	0.00	44.56	0.00	58.27	0.00
BMI (kg/m ²)	26.55±3.18	24.04±3.18	26.57±3.24	25.18±3.79	26.21±3.18	24.27±3.50	26.16±3.37	24.33±3.45	25.92±3.31	24.02±3.46
TC (mg/dl)	199.03±43.09	186.26±36.77	177.53±45.34	185.84±35.79	172.73±41.06	179.97±33.39	173±46.94	182.79±32.00	162.42±40.64	173.93±33.35
HDL-C (mg/dl)	40.83±9.92	50.67±12.23	38.64±9.85	49.92±12.42	39.23±9.87	51.33±12.42	39.57±10.12	53.04±12.59	38.02±9.80	46.27±11.78
TG (mg/dl)	164.73±116.78	130.46±90.76	170.43±107.65	159.24±115.52	163.13±96.40	135.42±96.53	161.84±98.31	135.06±105.87	161.89±80.83	159.88±111.48
LDL-C (mg/dl)	125.04±39.74	109.76±32.26	99.55±37.93	105.58±30.35	94.78±32.56	102.33±28.40	96.35±33.21	104.15±27.53	91.21±32.74	97.41±27.79
FPG (mg/dl)	110.75±41.96	91.83±16.66	104.61±28.87	97.16±27.06	107.48±34.93	92.43±31.29	106.70±34.80	91.32±29.12	107.44±35.73	99.38±26.03

SBP (mmHg)	126.27±20.05	120.40±13.76	124.71±17.01	140.66±24.73	123.69±17.74	139.84±21.46	125.85±17.34	135.33±20.63	124.61±15.71	129.63±21.47
DBP (mmHg)	76.27±11.12	77.10±9.08	78.27±10.86	88.09±14.77	77.04±11.25	84.78±11.15	78.07±10.89	83.51±11.31	76.58±10.12	80.63±12.78
Hypertension (%)	55.05	11.46	56.14	61.33	56.24	52.73	58.72	35.00	59.52	31.90
Diabetes(%)	21.58	1.96	22.87	0.00	24.83	5.09	25.66	2.85	28.48	4.04
Cigarette Smoking (%)	70.69	54.85	68.32	32.74	68.19	41.40	60.96	40.61	61.36	59.32
Alcohol consumers (%)	50.10	42.90	46.83	24.71	45.02	37.87	43.65	39.66	45.23	27.66

Values with '±' are mean ± s.d. PCI, percutaneous coronary intervention; CABG, coronary artery bypass graft; MI, myocardial infarction; BMI, body mass index; TC, total cholesterol; HDL-C, high density lipoprotein cholesterol; TG, triglyceride; LDL-C, low density lipoprotein cholesterol; FPG, fasting plasma glucose; SBP, systolic blood pressure; DBP, diastolic blood pressure; Hypertension was defined as an average systolic BP of ≥ 140 mmHg, an average diastolic BP ≥ 90 mmHg, or current treatment for hypertension with antihypertensive medication.

Supplementary Table 2: Discovery, Replication 1 and Replication 2 results for SNPs selected for replication

SNP	Loci	Position	Allele	Freq	Discovery – BAS			Discovery – CAS			Discovery – BAS+CAS			Replication 1		Discovery + Replication1		Replication2		Combined		
					SNP Type	Rsq	P	SNP Type	Rsq	P ^a	P ^b	P ^c	Cases/ Controls	P	Cases/ Controls	P	Cases/ Controls	P	Cases/ Controls	P	Cases/ Controls	
SNPs showed potential associations from meta-analysis of the two GWA study (P<1.0×10 ⁻⁴)																						
rs493549	1p31.3	68005632	A	0.48	g		1.19E-02	i	0.98	8.13E-04	3.15E-05	3.94E-05	1509/4938	5.56E-01	8721/5123	1.06E-02	10230/10061					
rs581244	1p22.1	94306578	A	0.19				i	0.96	9.70E-06	9.70E-06	1.11E-04	956/3840	2.00E-01	8704/5131	4.48E-03	9660/8971					
rs11166588	1p21.2	101552262	C	0.20			6.43E-02	g		5.78E-04	9.45E-05	3.88E-05	1515/5011	7.89E-01	8753/5132	1.90E-02	10268/10143					
rs6540796	1q32.3	211888290	A	0.38				g		7.76E-05	7.76E-05	7.60E-04	1009/3992	2.66E-01	8709/5124	1.30E-02	9718/9116					
rs11117722	1q41	215143186	C	0.21	g			i	0.99	3.50E-05	3.50E-05	4.39E-05	1004/3970	4.00E-01	8736/5125	2.76E-01	9740/9095					
rs2167787*	2p25.1	7228587	T	0.15			3.61E-02	g		5.75E-03	5.73E-04	3.14E-03	1512/5011	3.72E-01	8743/5128	2.32E-02	10255/10139					
rs2123536	2p24.1	19809058	T	0.39				g		4.11E-06	4.11E-06	4.92E-05	1008/3982	1.56E-03	8732/5175	3.92E-06	9740/9157	9.56E-04	2373/1958	1.83E-08	12113/11115	
rs195608	2p16.2	53345882	T	0.13				g		8.69E-05	8.69E-05	4.43E-05	995/3753	5.43E-01	8717/5126	1.65E-02	9712/8879					
rs4954477	2q22.1	137656361	C	0.19	g			g		4.32E-06	4.32E-06	4.41E-06	1009/3996	3.89E-01	8749/5134	1.87E-01	9758/9130					
rs2075302	2q24.2	162784392	C	0.14				g		3.45E-05	3.45E-05	9.33E-05	1006/3989	1.02E-01	8649/5097	1.30E-03	9655/9086					
rs13031005	2q31.1	170515057	G	0.56		i	1.00	5.12E-01	i	1.00	1.15E-05	3.97E-05	9.97E-06	1509/4998	8.83E-01	8707/5129	1.96E-02	10216/10127				
rs6731238	2q32.3	196830738	T	0.73					g		6.90E-05	6.90E-05	5.20E-04	1002/3974	2.77E-01	8686/5129	6.17E-01	9688/9103				
rs117859753	3q12.1	101115265	A	0.11	g			g		4.94E-05	4.94E-05	5.69E-06	981/3754	1.74E-03	8731/5177	1.46E-06	9712/8931	1.10E-01	2394/2080	5.42E-07	12106/11011	
rs4698123	4p15.32	15449542	G	0.17				i	0.91	4.23E-05	4.23E-05	6.70E-05	958/3698	9.23E-02	8740/5134	8.39E-01	9698/8832					
rs17006191	4q21.22	83871118	T	0.19				g		6.15E-05	6.15E-05	9.67E-04	1002/3983	5.75E-01	8746/5131	4.72E-02	9748/9114					
rs13148112	4q28.3	138338355	G	0.13		i	0.99	1.31E-01	g		1.03E-04	3.96E-05	5.35E-05	1515/5016	1.22E-01	8701/5129	4.94E-01	10216/10145				
rs1842896	4q32.1	156730909	T	0.76	i	1.00	3.61E-04	i	0.99	4.08E-03	1.79E-05	9.75E-05	1503/4991	5.79E-05	8644/5158	6.56E-08	10147/10149	2.97E-03	2324/1961	7.25E-10	12471/12110	
rs117297784	5p15.2	14539307	G	0.08	g			g		6.94E-05	6.94E-05	5.60E-04	980/3761	3.95E-05	8628/5148	1.49E-07	9608/8909	1.65E-01	2384/1987	1.13E-07	11992/10896	
rs4074793	5q11.2	52228882	G	0.07			6.29E-02	i	1.00	2.17E-04	3.64E-05	8.69E-05	1515/5012	1.29E-01	8709/5129	8.51E-04	10224/10141					
rs9349379	6p24.1	13011943	G	0.74				g		8.55E-05	8.55E-05	7.01E-05	1007/3964	1.58E-03	8636/5159	5.63E-06	9643/9123	2.41E-05	2346/2069	1.76E-09	11989/11192	
rs62402640	6p22.3	19841762	A	0.93				g		2.28E-05	2.28E-05	2.47E-04	995/3769	1.85E-01	8741/5130	8.10E-01	9736/8899					
rs9268402	6p21.32	32449331	G	0.59	g		1.26E-02	g		1.47E-03	6.12E-05	6.97E-04	1510/5009	3.32E-11	8708/5120	9.92E-14	10218/10129					
rs3734312	6q14.1	76717117	T	0.75	g			g		6.57E-05	6.57E-05	3.07E-04	1009/3997	2.98E-01	8625/5100	5.33E-01	9634/9097					
rs58205153	6q15	90708695	A	0.19				g		3.86E-05	3.86E-05	3.32E-04	990/3767	1.17E-01	8728/5128	8.78E-01	9718/8895					
rs4709988	6q27	166151767	C	0.62		g	5.08E-06	g		2.41E-03	7.99E-07	5.28E-05	1507/4985	8.77E-01	8705/5124	5.99E-02	10212/10109					
rs11988224	8p23.2	3060129	C	0.61				g		3.37E-05	3.37E-05	1.94E-04	1007/3987	4.47E-02	8679/5127	6.04E-04	9686/9114					
rs9632884	9p21.3	22062301	C	0.71	g		7.88E-04	i	0.94	2.26E-03	7.04E-06	2.87E-06	1450/4744	5.21E-29	8387/5158	1.41E-33	9837/9902					
rs10757274	9p21.3	22086055	G	0.46	i	0.99	4.70E-03	i	0.99	3.05E-05	1.66E-07	5.76E-08	1500/4954	1.15E-39	8691/5134	7.56E-45	10191/10088					
rs1333042	9p21.3	22093813	G	0.67	g		1.72E-05	i	0.99	425E-04	6.1E-08	5.06E-08	1479/4918	6.06E-34	8612/5141	3.22E-40	10091/10059					

rs1333049	9p21.3	22115503	C	0.49	g		6.37E-04	g		2.33E-04	4.81E-07	2.63E-08	1508/5002	2.34E-34	8758/5174	1.74E-40	10266/10176						
rs700965	9q22.32	97550959	T	0.15	g		1.19E-01	i	0.97	4.85E-05	1.78E-05	1.35E-04	1474/4903	8.77E-02	8653/5111	7.36E-04	10127/10014						
rs12416202	10p14	8607895	C	0.15				i	0.97	8.73E-05	8.73E-05	4.66E-04	984/3879	2.79E-01	8681/5127	1.13E-02	9665/9006						
rs7093239	10p13	13941646	T	0.13				g		1.34E-06	1.34E-06	6.29E-07	994/3771	5.03E-01	8562/5111	9.87E-02	9556/8882						
rs4747695	10p11.23	30318242	T	0.12	i	1.00	4.19E-02	g		6.39E-05	7.59E-06	1.96E-05	1511/5015	7.73E-01	8727/5133	1.45E-02	10238/10148						
rs11037662	11p11.2	43807884	G	0.38	g		6.69E-03	g		1.53E-03	3.99E-05	1.75E-04	1511/5007	3.97E-01	8726/5128	8.42E-03	10237/10135						
rs11231640	11q13.1	63447625	T	0.88	i	0.94	6.36E-03	i	0.91	2.00E-03	5.13E-05	2.73E-04	1449/4687	1.34E-02	8678/5117	9.00E-05	10127/9804						
rs10891129	11q22.3	109869203	A	0.72				i	0.99	4.64E-05	4.64E-05	4.02E-04	1005/3967	2.30E-02	8739/5125	3.64E-04	9744/9092						
rs7136259	12q21.33	88605319	T	0.39	g		3.30E-02	g		1.21E-03	1.08E-04	1.95E-05	1502/4993	4.73E-04	8732/5173	2.39E-07	10234/10166	5.27E-02	2394/2058	4.48E-08	12628/12224		
rs11066280	12q24.13	111302166	A	0.17	g		5.50E-01	g		3.00E-07	2.45E-06	1.04E-06	1478/4885	2.00E-06	8744/5169	4.92E-11	10222/10054	4.92E-02	2396/2069	1.69E-11	12618/12123		
rs876167	13q12.13	24613366	A	0.81				g		1.47E-05	1.47E-05	2.42E-05	1004/3973	3.23E-01	8691/5118	3.93E-01	9695/9091						
rs9599936	13q21.33	71585964	A	0.46				g		2.92E-05	2.92E-05	1.70E-05	993/3762	6.29E-02	8744/5125	3.60E-04	9737/8887						
rs1347331	13q33.2	103942694	T	0.11	i	0.99	1.89E-02	i	0.97	8.75E-04	4.91E-05	5.21E-06	1490/4954	4.53E-01	8731/5126	2.75E-03	10221/10080						
rs1998777	14q22.2	52365456	A	0.73	g		3.12E-05				3.12E-05	1.07E-04	505/1021	1.32E-01	8696/5125	7.78E-01	9201/6146						
rs11620677	14q24.2	71878184	A	0.20	i	0.99	1.53E-02	i	1.00	1.95E-03	9.56E-05	1.37E-04	1507/4993	6.80E-01	8718/5130	2.04E-02	10225/10123						
rs4778415	15q11.2	21964433	C	0.69	i	0.98	3.49E-02	i	0.97	1.27E-04	1.23E-05	1.57E-04	1451/4800	8.39E-01	8645/5095	8.74E-02	10096/9895						
rs12899034	15q14	33792022	A	0.62				g		3.34E-05	3.34E-05	1.32E-03	932/3655	4.24E-02	8609/5090	1.31E-03	9541/8745						
rs9652468	15q21.3	54611205	A	0.28				g		5.13E-05	5.13E-05	1.23E-02	1009/3996	5.37E-01	8736/5130	9.61E-02	9745/9126						
rs3825979	15q23	65468010	A	0.19	i	0.92	1.87E-01	i	0.96	1.45E-04	8.08E-05	2.08E-04	1449/4867	1.53E-01	8710/5126	1.84E-03	10159/9993						
rs77451922	16q21	56959062	T	0.67				g		5.28E-05	5.28E-05	1.54E-04	996/3769	3.91E-03	8648/5094	3.08E-01	9644/8863						
rs16955705	16q23.2	80230851	A	0.59	g		1.12E-02	g		2.36E-03	9.24E-05	3.16E-04	1508/5002	3.50E-01	8737/5126	9.08E-03	10245/10128						
rs11080326	17q12	30588522	C	0.50	i	0.98	1.12E-02	i	0.95	2.45E-03	9.68E-05	3.81E-03	1409/4624	7.06E-01	8745/5130	2.66E-01	10154/9754						
rs74504612	18q23	72002178	C	0.11				g		6.80E-05	6.80E-05	1.32E-03	996/3763	4.60E-01	8726/5117	3.51E-02	9722/8880						
rs4806229	19q12	33488450	C	0.83	i	0.98	1.64E-01	i	0.97	1.42E-04	6.81E-05	6.90E-05	1499/4950	9.33E-02	8748/5131	6.62E-04	10247/10081						
rs1201900	20q13.33	58704424	G	0.55	g		9.10E-01	g		3.86E-06	5.55E-05	7.69E-04	1484/4939	9.16E-02	8727/5127	1.67E-03	10211/10066						
rs17793491	20q13.33	58763696	C	0.71				i	0.91	2.14E-05	2.14E-05	5.97E-04	933/3602	7.35E-01	8643/5102	2.55E-01	9576/8704						
SNPs had a consistent association at $P \leq 1.0 \times 10^{-2}$ in both discovery populations																							
rs10493945	1p21.2	101538513	T	0.08	g		1.07E-03	g		9.88E-03	1.01E-04	4.57E-05	1515/5014	1.53E-01	8741/5129	9.60E-04	10256/10143						
rs2054120†	2p15	63765320	G	0.88	i	0.94	3.74E-04	i	0.96	6.08E-03	3.00E-05	6.47E-04	1484/4893										
rs6819096	4p14	36326499	A	0.38	i	0.99	9.01E-03	g		6.48E-03	2.39E-04	2.42E-03	1512/5010	1.90E-02	8692/5121	3.84E-04	10204/10131						
rs3775803*	4q31.22	146267573	G	0.88	i	0.99	7.43E-03	i	1.00	1.12E-02	3.89E-04	1.97E-03	1507/5010	1.47E-01	8744/5129	5.35E-03	10251/10139						
rs1181799	7q31.33	123962989	G	0.88	i	1.00	7.11E-03	i	0.98	7.66E-03	2.46E-04	1.67E-03	1514/5007	1.72E-01	8678/5132	7.74E-01	10192/10139						
rs296890	9q21.32	85785551	A	0.26	i	0.95	7.99E-03	i	0.96	8.14E-03	2.85E-04	2.48E-04	1401/4665	2.82E-01	8730/5130	5.76E-03	10131/9795						

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rs3826509	17p13.2	5323470	G	0.73	i	0.99	1.98E-02	i	0.99	1.40E-02	9.60E-04	4.23E-03	1484/4922	7.02E-01	8633/5115	2.68E-01	10117/10037
rs9897994	17q25.1	71216460	G	0.61	i	0.96	4.34E-04				4.34E-04	1.10E-04	480/1010	2.09E-01	8718/5118	1.84E-02	9198/6128
rs1156209	18q23	73692183	A	0.94	i	0.79	5.93E-01	g		1.49E-04	4.06E-04	5.56E-03	1480/4935	6.53E-02	8698/5128	4.97E-01	10178/10063

Alleles given are listed as risk allele. Freq, the frequency of risk allele based on discovery samples. SNP Type (g/i), the type of SNP (genotyped / imputed). Rsq, R-squared imputation quality value computed by the MACH software. The 11 SNPs at eight CAD loci identified in Chinese GWAS were represented in bold.

* For technical reasons in primer design, these SNPs were selected instead for replication.

†The SNPs did not pass the quality control in the replication 1 stage.

Note: In the discovery stage, we had initially genotyped 819 cases and 3529 controls of CAS, which were utilized to select 96 SNPs for replication. After the start of replication stage, additional 191 cases and 469 controls of CAS were genotyped using the Axiom™ Genome-Wide CHB 1 Array Plate and passed quality control. Then we reanalyzed our CAS GWAS data after the addition of these newly genotyped samples (In total, 1010 cases and 3998 controls).

P^a, P value of initial CAS GWAS data (819 cases and 3529 controls). P^b, the P value of discovery meta-analysis using initial CAS GWAS data (819 cases and 3529 controls) and BAS data (505 cases and 1021 controls). P^c, the P value of discovery meta-analysis using CAS GWAS data (1010 cases and 3998 controls) and BAS data.

Supplementary Table 3: Association results for four previously reported CAD loci in the present GWAS

SNP	Locus	Position	Risk Allele (Freq)	Nearby gene(s)	Discovery meta-analysis (BAS+CAS)			Replication1			Replication2			Combined Results (Discovery + Replication)		
					OR (95%CI)	P	Sample size (case/control)	OR (95%CI)	P	Sample size (case/control)	OR (95%CI)	P	Sample size (case/control)	OR (95%CI)	P	Sample size (case/control)
rs9349379	6p24.1	13011943	G(0.74)	<i>PHACTR1</i>	1.30 (1.14,1.47)	7.01E-05	1007/3964	1.10 (1.04,1.17)	1.58E-03	8636/5159	1.23 (1.12,1.36)	2.41E-05	2346/2069	1.15 (1.10,1.21)	1.76E-09	11989/11192
rs12524865	6q23.2	134238367	C(0.61)	<i>TCF21</i>	1.15 (1.04,1.25)	3.40E-03	1511/4990	1.11 (1.06,1.17)	8.46E-05	8615/5165	1.09 (1.00,1.19)	5.43E-02	2367/2076	1.11 (1.06,1.16)	1.87E-07	12493/12231
rs9632884	9p21.3	22062301	C(0.71)	<i>CDKN2A/B</i>	1.27 (1.15,1.41)	2.87E-06	1450/4744	1.39 (1.31,1.47)	5.21E-29	8387/5158				1.36 (1.29,1.43)	1.41E-33	9837/9902
rs10757274	9p21.3	22086055	G(0.46)	<i>CDKN2A/B</i>	1.28 (1.17,1.39)	5.76E-08	1500/4954	1.40 (1.33,1.47)	1.15E-39	8691/5134				1.37 (1.31,1.43)	7.56E-45	10191/10088
rs1333042	9p21.3	22093813	G(0.67)	<i>CDKN2A/B</i>	1.30 (1.19,1.44)	5.06E-08	1479/4918	1.40 (1.32,1.47)	6.06E-34	8612/5141				1.37 (1.31,1.44)	3.22E-40	10091/10059
rs1333049	9p21.3	22115503	C(0.49)	<i>CDKN2A/B</i>	1.28 (1.18,1.40)	2.63E-08	1508/5002	1.36 (1.30,1.43)	2.34E-34	8758/5174				1.34 (1.28,1.40)	1.74E-40	10266/10176
rs11066280	12q24.13	111302166	A(0.17)	<i>C12orf51</i>	1.32 (1.18,1.48)	1.04E-06	1478/4885	1.17 (1.10,1.26)	2.00E-06	8744/5169	1.11 (1.00,1.24)	4.92E-02	2396/2069	1.19 (1.13,1.25)	1.69E-11	12618/12123

Information for SNP ID and chromosomal position is based on NCBI genome build 36.

The risk allele frequency (Freq) in discovery studies is displayed. OR, odds ratio; CI, confidence interval

For each SNP, the odds ratios (with 95% confidence intervals) and *P* values represent the risk of CAD associated with each copy of the risk allele.

Supplementary Table 4: Effects of CAD loci on established cardiovascular risk factors

SNP	Chr	Position	Risk allele ^a	Lipid								Glucose		Obesity		Blood Pressure				Hypertension	
				HDL-C, mg/dl		LDL-C, mg/dl		TC, mg/dl		TG, mg/dl		FPG, mg/dl		BMI, kg/m ²		SBP, mmHg		DBP, mmHg		OR (95%CI)	P
				β (SE)	P	β (SE)	P	β (SE)	P	β (SE)	P	β (SE)	P	β (SE)	P	β (SE)	P	β (SE)	P		
rs2123536	2	19809058	T	0.10 (0.15)	5.1E-01	0.56 (0.34)	9.7E-02	0.72 (0.4)	7.1E-02	0.00 (0.01)	5.1E-01	-0.19 (0.34)	5.7E-01	-0.01 (0.04)	8.1E-01	0.27 (0.26)	2.9E-01	0.19 (0.14)	1.8E-01	1.04 (0.99-1.09)	1.1E-01
rs1842896	4	156730909	T	-0.38 (0.16)	1.8E-02	-0.02 (0.37)	9.6E-01	-0.05 (0.44)	9.2E-01	0.01 (0.01)	6.3E-02	0.21 (0.35)	5.4E-01	0.03 (0.04)	5.1E-01	0.63 (0.27)	2.0E-02	0.42 (0.15)	5.4E-03	1.08 (1.03-1.14)	4.3E-03
rs9349379	6	13011943	G	0.33 (0.19)	8.3E-02	-0.07 (0.44)	8.7E-01	-0.01 (0.52)	9.8E-01	-0.01 (0.01)	3.8E-01	0.10 (0.45)	8.3E-01	-0.02 (0.05)	7.8E-01	-0.95 (0.34)	4.8E-03	-0.12 (0.18)	5.1E-01	0.92 (0.85-0.98)	7.9E-03
rs9268402	6	32449331	G	-0.26 (0.15)	8.6E-02	0.04 (0.35)	9.1E-01	-0.18 (0.41)	6.6E-01	0.00 (0.01)	9.4E-01	0.43 (0.33)	1.8E-01	0.00 (0.04)	9.3E-01	-0.57 (0.25)	2.4E-02	-0.29 (0.14)	4.5E-02	0.94 (0.88-0.99)	1.4E-02
rs12524865	6	134238367	C	-0.11 (0.16)	5.1E-01	-0.43 (0.39)	2.7E-01	-0.62 (0.45)	1.7E-01	0.00 (0.01)	6.2E-01	-0.58 (0.36)	1.0E-01	-0.05 (0.05)	3.0E-01	0.23 (0.27)	4.1E-01	0.03 (0.15)	8.4E-01	1.06 (1.00-1.11)	6.2E-02
rs9632884	9	22062301	C	0.25 (0.19)	2.0E-01	-0.58 (0.46)	2.0E-01	-0.49 (0.53)	3.5E-01	-0.01 (0.01)	1.4E-01	-0.28 (0.41)	5.0E-01	-0.01 (0.05)	7.9E-01	-0.04 (0.32)	9.0E-01	0.13 (0.18)	4.7E-01	1.01 (0.94-1.08)	8.0E-01
rs10757274	9	22086055	G	0.10 (0.18)	5.8E-01	-0.14 (0.42)	7.5E-01	-0.62 (0.49)	2.1E-01	-0.01 (0.01)	7.3E-02	0.03 (0.38)	9.4E-01	-0.03 (0.05)	5.4E-01	-0.28 (0.29)	3.4E-01	0.00 (0.16)	9.8E-01	1.00 (0.94-1.07)	9.2E-01
rs1333042	9	22093813	G	0.29 (0.18)	1.1E-01	-0.25 (0.44)	5.6E-01	-0.32 (0.51)	5.3E-01	-0.01 (0.01)	9.6E-02	0.12 (0.39)	7.7E-01	-0.03 (0.05)	5.6E-01	0.24 (0.3)	4.3E-01	0.16 (0.17)	3.4E-01	1.04 (0.98-1.11)	2.3E-01
rs1333049	9	22115503	C	0.04 (0.18)	8.3E-01	-0.16 (0.42)	7.0E-01	-0.60 (0.48)	2.1E-01	-0.01 (0.01)	1.4E-01	-0.13 (0.37)	7.3E-01	-0.01 (0.05)	8.0E-01	-0.21 (0.29)	4.6E-01	0.03 (0.16)	8.5E-01	0.99 (0.93-1.05)	8.2E-01
rs7136259	12	88605319	T	0.25 (0.14)	8.4E-02	0.02 (0.33)	9.5E-01	0.05 (0.39)	9.0E-01	-0.01 (0.01)	1.8E-01	0.52 (0.31)	9.1E-02	-0.02 (0.04)	6.6E-01	-1.14 (0.24)	1.6E-06	-0.57 (0.13)	2.1E-05	0.89 (0.84-0.94)	4.7E-06
rs11066280	12	111302166	A	-1.22 (0.22)	1.4E-08	-0.10 (0.5)	8.5E-01	-2.19 (0.59)	1.9E-04	-0.03 (0.01)	9.6E-04	0.11 (0.47)	8.1E-01	-0.03 (0.06)	6.6E-01	-1.50 (0.35)	2.0E-05	-1.02 (0.2)	2.0E-07	0.87 (0.79-0.94)	2.5E-04

Information for SNP ID and chromosomal position is based on NCBI genome build 36. a. Risk allele for CAD trait.

CAD, coronary artery disease; Chr, chromosome; β(SE), effect size (standard error); OR(95%CI), odds ratio (95% confidence intervals); HDL-C, high density lipoprotein cholesterol; LDL-C, low density lipoprotein cholesterol; TC, total cholesterol; logTG, the logarithms of triglyceride; FPG, fasting plasma glucose; BMI, body mass index; SBP, systolic blood pressure; DBP, diastolic blood pressure.

Variables adjusted in the regression models are: sex, age for HDL-C, LDL-C, total cholesterol, TG (natural logarithms transformed), FPG and BMI; sex, age, age², and BMI for SBP and DBP; sex, age and BMI for Hypertension. For participants who were taking antihypertensive medication, we added 10 mm Hg to observed SBP values and 5 mm Hg to DBP values.

Supplementary Table 5: Derived allele frequency of ethnic-specific association loci on 12q24 in three ethnic groups

Ethnic specific association	SNP	Gene annotation	Position	Ancestral allele	Derived allele	DAF			Phenotype reported previously		
						CEU	CHB	JPT	CAD	BP	Other traits
East Asian											
	rs3782886	<i>BRAP</i>	110594872	A	G	0	0.15	0.24	CAD ^{1,2}		ALT ³
	rs4646776	<i>ALDH2</i>	110714402	G	C	0	0.16	0.23			
	rs671	<i>ALDH2</i>	110726149	G	A	0	0.16	0.23	CAD ¹	BP ⁴	lipids ⁴ , GGT ³ , MCHC ³ , ESSC ⁵
	rs2074356	<i>C12orf51</i>	111129784	C	T	0	0.09	0.23			HDL-C ⁶ , GGT ⁶ , WHR ⁷ , ESSC ⁸
	rs77768175	<i>C12orf51</i>	111220501	A	G	NA	NA	NA			
	rs11066280	<i>C12orf51</i>	111302166	T	A	0	0.17	0.24		BP ⁴	ALT ⁶ , ESSC ⁸
European											
	rs3184504	<i>SH2B3</i>	110368991	C	T	0.41	0	0	CAD ^{9,10}	BP ¹¹	PLT ¹⁰ , T1D ¹² , EOS ⁹
	rs4766578	<i>ATXN2</i>	110388754	A	T	0.42	NA	NA	CAD ¹⁰	BP ¹¹	PLT ¹⁰
	rs10774625	<i>ATXN2</i>	110394062	G	A	0.42	0	0	CAD ¹⁰	BP ¹¹	PLT ¹⁰
	rs653178	<i>ATXN2</i>	110492139	A	G	0.41	0	0	CAD ¹⁰	BP ^{11,13}	PLT ¹⁰
	rs11065987	<i>BRAP</i>	110556807	A	G	0.34	0	0	CAD ¹⁰		PLT ¹⁰ , lipids ¹⁴
	rs17696736	<i>C12orf30</i>	110971201	A	G	0.35	0	0	CAD ¹⁰		PLT ¹⁰ , T1D ¹⁵
	rs17630235	<i>TRAFD1</i>	111076069	G	A	0.33	0	0	CAD ¹⁰		PLT ¹⁰
	rs11066188	<i>C12orf51</i>	111095097	G	A	0.32	0	0	CAD ¹⁰		PLT ¹⁰
	rs11066301	<i>PTPN11</i>	111355755	A	G	0.35	0	0	CAD ¹⁰		PLT ¹⁰
	rs11066320	<i>PTPN11</i>	111390798	G	A	0.35	0	0	CAD ¹⁰		PLT ¹⁰

Information for SNP and position is based on NCBI genome build 36. The ancestral and derived allele status was obtained from the UCSC annotations. DAF, derived allele frequency, is obtained from HapMap Phase II repository. Genetic information of rs77768175 is obtained from 1000 Genomes.

rs3782886, rs4646776, rs671, rs2074356, rs77768175 and rs11066280 were in strong LD with each other ($r^2 = 0.95\sim 1.00$) and associated with CAD in the discovery ($P < 1.0 \times 10^{-4}$). CAD, coronary artery disease; ALT, alanine aminotransferase; BP, blood pressure; GGT, gamma glutamyl transferase; MCHC, mean corpuscular hemoglobin concentration; ESSC, esophageal squamous-cell carcinoma; HDL-C, high density lipoprotein cholesterol; WHR, waist hip ratio; PLT, platelet count; T1D, type 1 diabetes; EOS, eosinophil count.

Supplementary Table 6: Pairwise linkage disequilibrium coefficients between the four SNPs of 9p21

	rs9632884	rs10757274	rs1333042	rs1333049
rs9632884		0.27	0.65	0.29
rs10757274	NA		0.40	0.78
rs1333042	0.90	NA		0.43
rs1333049	0.84	NA	0.87	

Note: r^2 for HapMap CEU sample below diagonal, r^2 for control group of replication 1 above diagonal

Supplementary Table 7: Conditional logistic regression analysis for SNPs from 9p21.3

SNP	Position	Risk Allele	Frequency		Adjusted <i>P</i> Values ^a			
			CAD	Controls	rs9632884	rs10757274	rs1333042	rs1333049
rs9632884	22062301	C	0.76	0.70	NA	5.58E-08	0.02	1.46E-08
rs10757274	22086055	G	0.52	0.43	1.18E-18	NA	2.94E-13	1.44E-06
rs1333042	22093813	G	0.72	0.65	3.11E-09	3.71E-08	NA	7.96E-09
rs1333049	22115503	C	0.54	0.46	2.03E-15	0.10	2.68E-10	NA

a. The corresponding adjusted *P* values for each marker when the association is tested conditional on the observed association of each of the other markers.

Supplementary Table 8: Comparison of the four new findings between the present GWAS and the CARDIoGRAM datasets

SNP	CHR	BP	Risk allele	Other allele	Chinese population			European population		
					Our GWAS in combined-analysis (16975 cases/16491 controls)			CARDIoGRAM datasets (22233 cases/64762 controls)		
					Risk Allele Frequency	OR (95% CI)	<i>P</i>	Risk Allele Frequency	OR (95% CI)	<i>P</i>
rs2123536	2	19809058	T	C	0.39	1.12 (1.08,1.16)	6.83E-11	0.05	1.10 (1.03,1.17)	3.80E-03
rs1842896	4	156730909	T	G	0.76	1.14 (1.10,1.19)	1.26E-11	0.49	1.01 (0.98, 1.04)	0.482
rs9268402	6	32449331	G	A	0.59	1.16 (1.12,1.20)	2.77E-15	0.53	0.98 (0.95, 1.00)	0.093
rs7136259	12	88605319	T	C	0.39	1.11 (1.08,1.15)	5.68E-10	0.42	1.03 (1.00, 1.06)	0.035

Supplementary Table 9: Comparison of the present GWAS in discovery stage with published GWAS for CAD

Supplementary Table 1: Comparison of the present GWAS in discovery stage with published GWAS for CHD																
Previous GWAS									Our Study in discovery							
Locus	Nearby gene	SNPs	Position	RA/ OA	RAF	OR	P value	Reference	SNPs	RA/ OA	RAF ^a	RAF ^b	OR	P value	Power	
Comparison the association results between published GWAS in European and our study in discovery stage																
Group1	1p32.2	PPAP2B	rs17114036	56735409	A/G	0.91	1.17	3.81E-19	Schunkert H. et al ¹⁶	rs17114036	A/G	0.97	0.98	1.64	1.43E-03	0.34
	1q41	MIA3	rs17465637	220890152	C/A	0.72	1.14	1.40E-09	Kathiresan S. et al ¹⁷	rs17465637	C/A	0.60	0.54	1.12	1.01E-02	0.92
	6p24.1	PHACTR1	rs12526453	13035530	C/G	0.65	1.12	1.30E-09	Kathiresan S. et al ¹⁷	rs12526453	C/G	NA	1.00	NA	NA	
	6q23.2	TCF21	rs12190287	134256218	C/G	0.62	1.08	1.07E-12	Schunkert H. et al ¹⁶	rs9349379 ^c	G/A	0.74	0.80	1.30	7.01E-05	
										rs12190287	C/G	0.64	0.59	1.13	3.10E-02	0.55
										rs12524865 ^d	C/A	0.61	0.63	1.15	3.40E-03	
	9p21.3	CDKN2A/B	rs10757274	22086055	G/A	0.49	1.29	3.70E-06	McPherson R. et al ¹⁸	rs10757274	G/A	0.46	0.46	1.28	5.76E-08	0.99
			rs4977574	22088574	G/A	0.56	1.29	2.70E-44	Kathiresan S. et al ¹⁷	rs4977574	G/A	0.46	0.46	1.27	6.61E-08	0.99
			rs10757278	22114477	G/A	0.45	1.28	1.20E-20	Helgadottir A. et al ¹⁹	rs10757278	G/A	0.49	0.47	1.28	3.34E-08	0.99
			rs1333049	22115503	C/G	0.47	1.36	2.91E-19	Samani N.J. et al ²⁰ WTCCC ¹⁵	rs1333049	C/G	0.49	0.48	1.28	2.63E-08	0.99
	10q23.31	LIPA	rs1412444	90992907	T/C	0.42	1.09	2.76E-13	Peden J.F. et al ²¹	rs1412444	T/C	0.33	0.37	1.12	4.75E-02	0.63
	10q24.32	CYP17A1-NT5C2	rs12413409	104709086	G/A	0.89	1.12	1.03E-09	Schunkert H. et al ¹⁶	rs12413409	G/A	0.73	0.74	1.13	1.89E-02	0.77
	11q22.3	PDGFD	rs974819	103165777	T/C	0.32	1.07	2.41E-09	Peden J.F. et al ²¹	rs974819	T/C	0.63	0.62	1.12	1.26E-02	0.47
	12q24.12	SH2B3	rs3184504	110368991	T/C	0.38	1.13	8.60E-08	Gudbjartsson D.F. et al ⁹	rs3184504	T/C	NA	0.00	NA	NA	
										rs11066280	A/T	0.17	0.17	1.32	1.04E-06	
	15q25.1	ADAMTS7	rs1994016	76867289	C/T	0.60	1.19	4.98E-13	Reilly M.P. et al ²²	rs1994016	C/T	0.93	0.90	1.28	1.20E-01	0.66
			rs3825807	76876166	A/G	0.57	1.08	1.07E-12	Schunkert H. et al ¹⁶	rs3825807	A/G	NA	0.89	NA	NA	0.31
			rs4380028	76898148	C/T	0.65	1.07	3.71E-09	Peden J.F. et al ²¹	rs4380028	C/T	0.62	0.58	1.17	6.67E-04	0.47
17p13.3	SMG6-SRR	rs216172	2073254	C/G	0.37	1.07	1.15E-09	Schunkert H. et al ¹⁶	rs216172	C/G	0.26	0.21	1.11	3.30E-02	0.42	
Group2	1p32.3	PCSK9	rs11206510	55268627	T/C	0.81	1.15	9.60E-09	Kathiresan S. et al ¹⁷	rs11206510	T/C	0.94	0.93	1.19	9.75E-02	0.46
	1p13.3	CELSR2-SORT1	rs599839	109623689	A/G	0.77	1.29	4.05E-09	Samani N.J. et al ²⁰	rs599839	A/G	0.93	0.97	1.16	3.75E-01	0.90
	2q33.1	WDR12	rs6725887	203454130	C/T	0.14	1.17	1.30E-08	Kathiresan S. et al ¹⁷	rs6725887	C/T	0.02	0.02	0.89	5.21E-01	0.30
	3q22.3	MRAS	rs9818870	139604812	T/C	0.15	1.15	7.44E-13	Erdmann J. et al ²³	rs9818870	T/C	NA	0.00	NA	NA	
										rs1199337 ^e	C/G	0.24	0.11	1.13	1.97E-02	
										rs17609940	G/C	0.97	0.96	0.95	6.74E-01	0.14
	6p21.31	ANKS1A	rs17609940	35142778	G/C	0.75	1.07	1.36E-08	Schunkert H. et al ¹⁶	rs3798220	C/T	NA	0.07	NA	NA	0.99
	6q26	LPA	rs3798220	160881127	C/T	0.02	1.92	9.00E-07	Clarke R. et al ²⁴	rs10455872	G/A	NA	0.00	NA	NA	
			rs10455872	160930108	G/A	0.07	1.70	3.40E-15	Clarke R. et al ²⁴	rs3125055 ^f	A/T	0.11	0.12	1.17	2.86E-02	
	7q32.2	ZC3HC1	rs11556924	129450732	C/T	0.62	1.09	9.18E-18	Schunkert H. et al ¹⁶	rs11556924	C/T	0.97	0.97	1.67	6.06E-02	0.17
	17p11.2	RASD1-PEMT	rs12936587	17484447	G/A	0.56	1.07	4.45E-10	Schunkert H. et al ¹⁶	rs12936587	G/A	NA	0.90	NA	NA	0.25
										rs9899364 ^g	A/G	0.42	0.46	1.15	8.10E-03	
	19p13.2	LDLR	rs1122608	11024601	G/T	0.75	1.15	1.90E-09	Kathiresan S. et al ¹⁷	rs1122608	G/T	0.90	0.90	0.79	8.54E-02	0.62
	21q22.11	MRPS6	rs9982601	34520998	T/C	0.13	1.20	6.40E-11	Kathiresan S. et al ¹⁷	rs9982601	T/C	NA	0.01	NA	NA	0.23
	Group3	7q22	BCAP29	rs10953541	107031781	C/T	0.80	1.08	3.12E-08	Peden J.F. et al ²¹	rs10953541	C/T	0.84	0.79	1.04	5.64E-01
9q34.2		ABO	rs579459	135143989	C/T	0.21	1.10	4.08E-14	Schunkert H. et al ¹⁶	rs579459	C/T	0.21	0.16	1.04	4.55E-01	0.60
10p11.23		KIAA1462	rs2505083	30375128	C/T	0.38	1.07	3.87E-08	Peden J.F. et al ²¹	rs2505083	C/T	0.24	0.31	0.96	4.97E-01	0.41

10q11.21	<i>CXCL12</i>	rs1746048	44095830	C/T	0.84	1.17	7.40E-09	Kathiresan S. et al ¹⁷	rs1746048	C/T	0.65	0.67	1.01	8.33E-01	0.97
11q23.3	<i>APOA5-A4-C3-A1</i>	rs964184	116154127	G/C	0.13	1.13	1.02E-17	Schunkert H. et al ¹⁶	rs964184	G/C	0.22	0.24	1.06	2.41E-01	0.80
13q34	<i>COL4A1-A2</i>	rs4773144	109758713	G/A	0.44	1.07	3.84E-09	Schunkert H. et al ¹⁶	rs4773144	G/A	0.38	0.43	1.03	7.16E-01	0.48
14q32.2	<i>HHIPL1</i>	rs2895811	99203695	C/T	0.43	1.07	1.14E-10	Schunkert H. et al ¹⁶	rs2895811	C/T	0.27	0.27	0.98	6.82E-01	0.43
17q21.32	<i>UBE2Z</i>	rs46522	44343596	T/C	0.53	1.06	1.81E-08	Schunkert H. et al ¹⁶	rs46522	T/C	0.72	0.79	1.07	1.57E-01	0.35
Comparison the association results between published GWAS in Chinese and our study in discovery stage															
6p24.1	<i>C6orf105</i>	rs6903956	11882569	A/G	0.07	1.51	4.87E-12	Wang F. et al ²⁵	rs6903956	A/G	0.08	0.04	0.98	8.15E-01	0.99

Information for SNP ID and chromosomal position is based on NCBI genome build 36. Power is estimated for test of CAD association in 1515 cases and 5019 controls at a significance level of $P < 0.05$ with OR reported in Europeans and risk allele frequency in Chinese. RA, risk allele; OA, other allele; RAF, risk allele frequency; NA, no available.

a. risk allele frequency in our discovery samples; b. risk allele frequency in HapMap CHB data; c. r^2 between rs12526453 and rs9349379 is 0.30; d. r^2 between rs12190287 and rs12524865 is 0.61; e. r^2 between rs9818870 and rs1199337 is 1.00; f. r^2 between rs10455872 and rs3125055 is 0.54; g. r^2 between rs12936587 and rs9899364 is 0.62. All the LD data with r^2 are obtained from HapMap phase II in CEU.

Supplementary Note: The CARDIoGRAM Consortium

The CARDIoGRAM Consortium

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